

Annexure B

List of projects

1. Viscosity measurement of liquids using viscometer
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3. Drag measurement for external flow over objects
4. Characterization of boundary layers
5. Major and minor losses in piping systems
6. Water sprinkler and conservation of angular momentum
7. Liquid sloshing in a tank subjected to rectilinear acceleration
8. Thermal management of devices using convective air flow
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10. Flow visualization (e.g. flow over a circular cylinder – dye injection in viscous fluid)
11. Pressure field across curved streamlines
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14. Flow evolution in jets, mixing layers, and wakes
15. Design and demonstration of pumps, water turbines
16. Building a model of piping systems for household/commercial application and demonstration

List of projects (ME 207)

1. Viscosity measurement of liquids using viscometer

Goals:

To build a viscometer for measuring viscosities of liquids

Broad expectations:

1. Viscosities should be measured for various liquids
2. For the same liquid, multiple experiments need to be conducted to measure viscosity and uncertainties need to be quantified.
3. The parameters in the experiments need to be identified and experiments need to be conducted for various values of these parameters.
4. Obtained values must be compared with the data in the literature.

2. Measurement of hydraulic jump radius

Goals:

To build an experimental setup for measuring hydraulic jump radius of liquid jets on a flat surface, as a function of nozzle height.

Broad expectations:

1. The hydraulic jump radius should be measured for at least three different nozzle heights, and a fixed flow rate.
2. For the same nominal conditions (i.e., nozzle height, flow rate etc.), multiple experiments need to be conducted to measure the jump radius.
3. Error and uncertainty quantification must be carried out for each experimental measurement.

3. Drag measurement for external flow over objects

Goals:

To study external air flow over objects and measure drag forces exerted by the object on the air flow.

Broad expectations:

1. Drag forces must be measured for different velocities, shapes, and sizes of objects. Some of the shapes that can be considered are spheres, cylinders, wings/airfoils etc.
2. For the same experimental conditions (velocity, size and shape of object), multiple experiments need to be conducted to measure drag force and uncertainties need to be quantified.
3. A correlation between drag coefficient and Reynolds number need to be developed. The obtained correlation needs to be compared with those available in the literature.

4. Characterization of boundary layers

Goals:

To build an experimental setup for measuring the axial velocity profile inside the Boundary layer over a flat plate.

Broad expectations:

1. A sufficiently thick boundary layer must be set up using a flow over a large flat plate.
2. A mechanism must be built to collect velocity data at multiple locations (at least 10 locations) within a given cross section of the boundary layer. Measurements may be taken in both laminar and turbulent parts of the boundary layer.
3. A comparison with theoretical values, wherever possible, must be carried out.
4. Error and uncertainty quantification must be carried out for each experimental measurement.

5. Major and minor losses in piping systems*

Goals:

To building an experimental setup for quantify both major and minor losses in piping systems

Broad expectations:

1. The experimental setup needs to be built from scratch
2. Major losses should be quantified at least for three different pipe diameters
3. Minor losses should be quantified at least for three different components such as valve, elbow, joints.
4. Data should also be recorded for multiple flow rates

5. For the same case, multiple readings must be taken, and uncertainties need to be quantified.
6. Recorded data must be compared with existing correlations.

6. Water sprinkler and conservation of angular momentum

Goals:

To build an experimental setup for testing the conservation of angular momentum using a water sprinkler.

Broad expectations:

1. Angular velocity of the water sprinkler should be measured for multiple flow rates.
2. Principles of angular momentum conservation must be applied to estimate the theoretical values of the angular velocity of the sprinkler as a function of flow rate. The experimental measurement and the theoretical estimates should be compared directly.
3. Error and uncertainty quantification must be carried out for each experimental measurement.

7. Liquid sloshing in a tank subjected to rectilinear acceleration

Goals:

To investigate liquid sloshing in a partially filled tank subject to rectilinear/~~circular~~ acceleration and to study the relationship between free surface orientation and acceleration

Broad expectations:

1. Orientation of free surface needs to be measured for various magnitudes and directions of accelerations (Note: accelerations need to be precisely controlled).
2. Determine if properties of liquid such as density, viscosity affect the orientation of free surface
3. Multiple experiments need to be conducted, and uncertainties need to be quantified.
4. The experimental results would need to be compared with analytical solution.

8. Thermal management of devices using convective air flow

Goals:

To build an experimental set up for cooling a heat source using air flow.

Broad expectations:

1. The setup should be able to cool a heat source (such as a heated plate) to steady state.
2. Air flow rate must be varied to achieve the steady state. One example is to try to mimic a desktop CPU.
3. Temperature measurements should be done at multiple locations within the flow.
4. Steady state temperature vs flow rate will be of interest to the community.
5. Safety and precaution must be taken before carrying out the experiment.

9. Design and demonstration of atomizer

Goals:

To building an atomizer that atomizes liquid stored in a container.

Broad expectations:

1. An atomizer must be built that creates a spray using liquid stored in a container based on the principle of Bernoulli's equation.
2. The rate of fluid entrainment and atomization needs to be measured as a function of air flow rate.
3. The pressure variation along the atomizer needs to be measured
4. Atomization experiments can be conducted for different fluids
5. Measurements must be compared with the analytical solution.

10. Flow visualization

Goals:

To observe flow patterns in one of the standard flows (such as flow around a circular cylinder) using dye injection.

Broad expectations:

1. The experimental setup should provide a clear visualization of streamlines.
2. Multiple experiments must be carried out at various Reynolds number.
3. The working liquid and dye (if required) must be chosen carefully.
4. Uncertainties in measurements must be carried out for each experimental reading.

11. Pressure field across curved streamlines

Goals:

To study the variation of pressure across curved streamlines in a fluid flow.

Broad expectations:

1. The experimental setup must be designed to set up a curved flow and then measure the pressure variations across curved streamlines.
2. The setup should examine the effect of streamline curvature on the pressure distribution.
3. Results should be compared with theoretical predictions.

12. Lift measurement for external flow over aircraft wing

Goals:

To measure lift forces on an airfoil at preferably different angles of attack.

Broad expectations:

1. Measure lift for various Reynolds number (flow velocities) and angle of attack.
2. Compare lift coefficients with theoretical values.
3. Errors and uncertainty quantification must be carried out for each experimental measurement.

13. Design and demonstration of vertical axis mini wind turbine

Goals:

To design and fabricate a vertical axis wind turbine for energy generation.

Broad expectations:

1. The setup should demonstrate the ability of the wind turbine to perform work and further assessing its performance under various operating conditions.
2. Test the wind turbine output at various wind speeds.
3. A pulley system housed on the shaft may demonstrate the ability to deliver work.
4. Compare your results with basic understanding of wind energy concepts.

14. Flow evolution in jets or mixing layers or wakes

Goals:

To study the development of velocity profile in jets or mixing layers or wakes.

Broad expectations:

1. The experimental setup should provide a clear picture of the evolving velocity field.
2. Measure velocity profiles at various cross sections.
3. Quantification of uncertainties in velocity measurements must be carried out.
4. A comparison with theoretical values should be conducted whenever possible.

15. Design and demonstration of water turbines

Goals:

To design and demonstrate the working principles of water turbines.

Broad expectations:

1. A small-scale water turbine (such as Pelton wheel) must be designed and demonstrated.
2. The experimental setup should be able to deliver power output.
3. A pulley system housed on the shaft may demonstrate the ability to deliver work.
4. Test the power delivered for varying flow rates of inflow.

16. Building a model of piping systems for household/commercial application and demonstration*

Goals:

To build an experimental set up for demonstrating the capability of supply and control of water flow rates in a replica of pipe networks in any household application.

Broad expectations:

1. Identify a moderately complicated pipe network used in any household/commercial buildings. One option is to choose the piping system used for district cooling at IIT Gandhinagar.
2. A lab-scale replica of the identified network must be built and connected to a water reservoir.
3. Demonstrate that the built set up is able to supply and control the water flow rate through various sections of the network as per requirement.
4. Detailed calculations, showing the choice of pipe diameter, head loss estimates, pumping power, etc. must be provided.